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Semiconductor structure and manufacturing method thereof

Abstract

The invention provides a semiconductor structure, which comprises a substrate, a first medium/high voltage device region and a low voltage device region are defined on the substrate, the first medium/high voltage device region comprises a first gate contact and a first source/drain contact, and the low voltage device region comprises two second source/drain contacts and a second gate contact, wherein the second gate contact is located between the two second source/drain contacts and directly contacts the two second source/drain contacts. A first conductive layer located in the first medium/high voltage device region, wherein the first conductive layer does not extend into the low voltage device region, and a second conductive layer located above the first conductive layer and spanning the first medium/high voltage device region and the low voltage device region.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The invention relates to the field of semiconductor manufacturing, in particular to a structure generated at the boundary of a medium/high voltage device region and a low voltage device region in the semiconductor manufacturing.

2. Description of the Prior Art

[0002] In the semiconductor process, the fineness of the nano-process is generally described in nanometers. For example, the 14-nanometer process means that the lowest line width that can be formed in the semiconductor process is 14 nanometers. With the progress of process technology, the line width of nano-process is gradually reduced.

[0003] However, even with the progress of nano-fabrication, not all devices are suitable for high-precision nano-fabrication. For example, when the size of the device does not match the precision of the nano-process, it will not only lead to the decline of the yield of the device, but also consume more costs. Therefore, in order to adapt to components with different sizes or precisions, different components will be formed by nano-processes with different precisions.

[0004] On one chip, there may be components with different precisions in different regions at the same time, which may be formed by nano-processes with different precisions. Therefore, at the boundary of these regions, various structural problems may easily occur.

SUMMARY OF THE INVENTION

[0005] The invention provides a semiconductor structure, which comprises a substrate, wherein a first medium/high voltage device region and a low voltage device region are defined on the substrate, the first medium/high voltage device region comprises a first gate contact and a first source/drain contact, and the low voltage device region comprises two second source/drain contacts and a second gate contact, wherein the second gate contact is located between the two second source/drain contacts and directly contacts the two second source/drain contacts. A first conductive layer located in the first medium/high voltage device region and electrically connected to the first gate contact or the first source/drain contact, wherein the first conductive layer does not extend into the low voltage device region, and a second conductive layer located above the first conductive layer and spanning the first medium/high voltage device region and the low voltage device region, wherein the second conductive layer is electrically connected with the first conductive layer.

[0006] The invention also provides a method for manufacturing a semiconductor structure, which comprises providing a substrate, wherein a first medium/high voltage device region and a low voltage device region are defined adjacent to each other, the first medium/high voltage device region comprises a first gate contact and a first source/drain contact, and the low voltage device region comprises two second source/drain contacts and a second gate contact, wherein the second gate contact is located between the two second source/drain contacts and directly contacts the two second sources/drains. Forming a first conductive layer located in the first medium/high voltage device region and electrically connected to the first gate contact or the first source/drain contact, wherein the first conductive layer does not extend into the low voltage device region, and forming a second conductive layer located above the first conductive layer and spanning the first medium/high voltage device region and the low voltage device region, wherein the second

conductive layer is electrically connected with the first conductive layer.
[0007] The applicant found that with the development of technology, the size of semiconductor devices is getting smaller and smaller, especially at the boundary of medium/high voltage device regions and low voltage device regions, various structural problems are easy to occur. For example, in the conventional 22 nm or 28 nm process, time dependent dielectric breakdown will not occur at the boundary of the medium/high voltage device region and the low voltage device region. However, as the process progresses below 17 nm, the conductive layer in the medium/high voltage device region is too close to the device in the low voltage device region, which may affect the quality of other devices due to time dependent dielectric breakdown. Therefore, the present invention aims at solving this problem, and specifically designs a semiconductor structure so that the first conductive layer in the medium/high voltage device region does not extend to the low voltage device region, and the second conductive layer in the medium/high voltage device region extends to the low voltage device region. Therefore, without changing the process conditions and being limited by the process, the problem of time dependent dielectric breakdown caused by the short distance between the conductive layer in the medium/high voltage device region and the device in the low voltage device region can be avoided, and the quality of the device can be improved.
[0008] These and other objectives of the present invention will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the preferred embodiment that is illustrated in the various figures and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] In order to make the following easier to understand, readers can refer to the drawings and their detailed descriptions at the same time when reading the present invention. Through the specific embodiments in the present specification and referring to the corresponding drawings, the specific embodiments of the present invention will be explained in detail, and the working principle of the specific embodiments of the present invention will be expounded. In addition, for the sake of clarity, the features in the drawings may not be drawn to the actual scale, so the dimensions of some features in some drawings may be deliberately enlarged or reduced.
[0010] FIG. **1** shows a schematic top view near a boundary between a medium/high voltage device region and a low-voltage device region in a first embodiment of the present invention.
[0011] FIG. **2** is a schematic cross-sectional view of the medium/high voltage device region and the low voltage device region in the first embodiment of the present invention, especially the cross-sectional structure taken along the cross-sectional line A-A′ in FIG. **1**.
[0012] FIG. **3** shows a schematic top view near a boundary between a medium/high voltage device region and a low-voltage device region in a second embodiment of the present invention.
[0013] FIG. **4** is a schematic cross-sectional view of the medium/high voltage device region and the low voltage device region in the second embodiment of the present invention, especially the cross-sectional structure taken along section line B-B′in FIG. **3**.
DETAILED DESCRIPTION
[0014] To provide a better understanding of the present invention to users skilled in the technology of the present invention, preferred embodiments are detailed as follows. The preferred embodiments of the present invention are illustrated in the accompanying drawings with numbered elements to clarify the contents and the effects to be achieved.
[0015] Please note that the figures are only for illustration and the figures may not be to scale. The scale may be further modified according to different design considerations. When referring to the words “up” or “down” that describe the relationship between components in the text, it is well

known in the art and should be clearly understood that these words refer to relative positions that can be inverted to obtain a similar structure, and these structures should therefore not be precluded from the scope of the claims in the present invention.
[0016] Although the present invention uses the terms first, second, third, etc. to describe elements, components, regions, layers, and/or sections, it should be understood that such elements, components, regions, layers, and/or sections should not be limited by such terms. These terms are only used to distinguish one element, component, region, layer and/or block from another element, component, region, layer and/or block. They do not imply or represent any previous ordinal number of the element, nor do they represent the arrangement order of one element and another element, or the order of manufacturing methods. Therefore, the first element, component, region, layer or block discussed below can also be referred to as the second element, component, region, layer or block without departing from the specific embodiments of the present invention.
[0017] The term “about” or “substantially” mentioned in the present invention usually means within 20% of a given value or range, such as within 10%, or within 5%, or within 3%, or within 2%, or within 1%, or within 0.5%. It should be noted that the quantity provided in the specification is approximate, that is, the meaning of “about” or “substantially” can still be implied without specifying “about” or “substantially”.
[0018] The terms “coupling” and “electrical connection” mentioned in the present invention include any direct and indirect means of electrical connection. For example, if the first component is described as being coupled to the second component, it means that the first component can be directly electrically connected to the second component, or indirectly electrically connected to the second component through other devices or connecting means.
[0019] Although the invention of the present invention is described below by specific embodiments, the inventive principles of the present invention can also be applied to other embodiments. In addition, in order not to obscure the spirit of the present invention, specific details are omitted, and the omitted details are within the knowledge of those with ordinary knowledge in the technical field.
[0020] Please refer to FIG. **1** and FIG. **2**, in which FIG. **1** shows a schematic top view near a boundary between a medium/high voltage device region and a low voltage device region in the first embodiment of the present invention, and FIG. **2** shows a schematic cross-sectional view near a boundary between a medium/high voltage device region and a low voltage device region in the first embodiment of the present invention, especially the cross-sectional structure taken along section line A-A′ in FIG. **1**. As shown in FIG. **1** and FIG. **2**, the semiconductor structure **10** of the present invention includes a medium/high voltage device region **10**, a low voltage device region **20** and another medium/high voltage device region **30**. The low voltage device region **20** is located between the medium/high voltage device region **10** and the medium/high voltage device region **30**. There is a boundary line B**1** between the medium/high voltage device region **10** and the low voltage device region **20**, and a boundary line B**2** between the medium/high voltage device region **30** and the low voltage device region **20**. Among them, the medium/high voltage device region **10**, the medium/high voltage device region **30** and the low voltage device region **20** are different in that the operating voltages of the elements or devices comprised therein are different. Generally speaking, taking a display chip as an example, the low voltage device region **20** includes, for example, a logic operation circuit, and its operating voltage is below 5 volts, preferably within 1.5 volts. On the contrary, the operating voltage of the electronic components comprised in the medium/high voltage device region **10** and the medium/high voltage device region **30** is more than 5 volts, usually more than 10 volts. For example, a driving element in a display chip needs a higher voltage to drive the elements therein, and these elements belong to the medium/high voltage elements of the present invention.
[0021] In this embodiment, the medium/high voltage device region **10** and the medium/high voltage device region **30** may have the same or similar structures. Take FIG. **1** as an example, the

medium/high voltage device region **10** and the medium/high voltage device region **30** have mutually symmetrical structures, and the elements or devices in them have the same materials and manufacturing methods, so only the medium/high voltage device region **10** and the low voltage device region **20** are mainly described in the following paragraphs, and the description of the medium/high voltage device region **30** is not repeated because it has the same characteristics as the medium/high voltage device region **10**.

[0022] As mentioned in the prior art, different devices on the same chip may be formed by different precision nano-processes because of their different sizes or applications. In this embodiment, the medium/high voltage devices or other related electronic devices (such as wires or trenches) in the medium/high voltage device region **10** are formed by a process step of 17 nm, for example. On the other hands, various low voltage devices (such as logic operation circuits) in the low voltage device region **20** require a higher nanometer precision, for example, by a process of 14 nm. However, it should be noted that the above 17 nm process and 14 nm processes are only some examples of the present invention, and the present invention is not limited to this.

[0023] When the nano-precision of semiconductor process is improved, it means that the size of devices is getting smaller and smaller, and devices can also develop towards three-dimensional structure to increase the density per unit area. In this embodiment, the low voltage device region **20** comprises a plurality of fin structures F to form a three-dimensional structure to reduce the device size. However, the medium/high voltage device region **10** needs to withstand high operating voltage, and the three-dimensional structure such as fin structure is easy to penetrate under high operating voltage, so the plane structure is mainly formed in the medium/high voltage device region **10**. That is to say, the top view of the semiconductor structure shown in FIG. **1** can be regarded as the boundary between two regions respectively comprising planar electronic components and regions comprising three-dimensional electronic components, wherein the two regions are formed by different nano-precision processes.

[0024] Referring to FIG. **1** and FIG. **2**, the medium/high voltage device region **10** includes medium/high voltage devices, such as a substrate **1**, on which there are an interlayer dielectric layer **12**, a diffusion region D, a gate structure G**1**, a gate contact MP**1** and source/drain contacts MD**1**. The low voltage device region **20** also includes various devices, such as the substrate **1**, the interlayer dielectric layer **12**, the fin structures F, the gate structure G**2**, the gate contact MP**2** and the source/drain contacts MD**2**. As shown in FIG. **2**, the gate structure G**1**, the gate structure G**2**, the gate contact MP**1**, the gate contact MP**2**, the source/drain contact MD**1** and the source/drain contact MD**2** are all located in an interlayer dielectric layer **12**. The material of the interlayer dielectric layer **12** is, for example, silicon oxide, silicon nitride or silicon oxynitride, but not limited thereto. In addition, in this embodiment, the top surfaces of the gate contact MP**1** and the source/drain contacts MD**1** located in the medium/high voltage device region **10** are aligned with the top surfaces of the gate contact MP**2** and the source/drain contacts MD**2** located in the low voltage device region **20**, but the present invention is not limited to this.

[0025] In more detail, taking the formation of high voltage transistor as an example, as shown in FIGS. **1** and **2**, high voltage transistors are formed in the medium/high voltage device region **10** and the medium/high voltage device region **30**, and these two high voltage transistors can be regarded as two independent elements. The low voltage device region **20** between the medium/high voltage device region **10** and the medium/high voltage device region **30** can be used as the peripheral region of the medium/high voltage device region to accommodate some dummy patterns, reduce exposure problems caused by pattern density differences, or form low voltage electronic elements in the low voltage device region **20**, all of which are within the scope of the present invention.

[0026] In this embodiment, the substrate **1** is, for example, a silicon substrate, and the diffusion region D in the medium/high voltage device region **10** and the fin structure F in the low voltage device region **20** are made of, for example, silicon, which are part of the substrate **1**. The diffusion

region D is a planar structure, while the fin structure F is a three-dimensional structure. The gate structures G**1** and G**2**, such as polysilicon gates or metal gates, span the diffusion region D and the fin structure F, and form various semiconductor devices. Taking this embodiment as an example, the gate structure G**1** spans the diffusion region D in the medium/high voltage device region **10** to form a high voltage transistor structure. In addition, in this embodiment, the gate structure G**2** also spans the fin structure F in the low voltage device region **20**. As mentioned above, the gate structure G**2** formed in the low voltage device region **20** can be used as the dummy pattern of the gate structure G**1** of the medium/high voltage device region **10** to reduce the pattern density difference between different regions. Or in other embodiments, the gate structure G**2** in the low voltage device region **20** can be used to form other electronic devices instead of being used as the dummy pattern of the gate structure G**1**, which is also within the scope of the present invention.
[0027] The materials of the gate contact MP**1**, the gate contact MP**2**, the source/drain contact MD**1** and the source/drain contact MD**2** are, for example, metals such as tungsten, cobalt, copper, aluminum, gold, silver, etc, but not limited thereto. The gate contact MP**1** is located in the medium/high voltage device region **10**, and the gate contact MP**2** is located in the low voltage device region **20**, which are used to electrically connect the gate structures G**1** and G**2** and other subsequent wires, so the gate contacts MP**1** and MP**2** are located above the gate structures G**1** and G**2** respectively. The source/drain contacts MD**1** are located on the diffusion region D for connecting the source/drain region of the high voltage transistor, and the source/drain contacts MD**2** are located on the fin structure F. The gate contacts MP**1** and MP**2** and the source/drain contacts MD**1** and MD**2** are used to connect other circuit layers formed subsequently.
[0028] It is worth noting that as the size of semiconductor devices is getting smaller and smaller, the sizes of the above-mentioned devices including gate structures G**1** and G**2**, gate contacts MP**1** and MP**2**, and source/drain contacts MD**1** and MD**2** are also reduced. In this case, the difficulty of the overlapping alignment step of the gate contact MP**1** and the gate structure G**1** will also increase. In other words, since the gate contact MP**1** needs to be accurately overlapped with the gate structure G**1** to be electrically connected with each other, as the sizes of the gate contact MP**1** and the gate structure G**1** become smaller and smaller, this overlapping step will become more and more difficult. On the other hand, the size of the gate contact MP**1** (horizontal dimension in cross section) is close to the exposure limit of the current machine, so it is difficult to make the gate contact MP**1** to a sufficient height, because this will lead to an excessive aspect ratio of the gate contact MP**1**, resulting in manufacturing defects (for example, the gate contact MP**1** is broken due to the difficulty in gap filling). In this embodiment, the height of the gate contact MP**1** can only be made to about 450 angstroms under the process of 17 nm. If the height of the gate contact MP**1** is too high, it will easily lead to the above-mentioned problem that the gate contact MP**1** is broken due to insufficient gap filling due to too large aspect ratio. Explain with the actual process, in other processes with larger dimensions, such as 22 nm or 28 nm, after the gate contact MP**1** is formed on the gate structure G**1**, the first conductive layer M**1** can be directly formed to connect with the gate contact MP**1**. However, when the process is reduced to less than 17 nm, the height of the gate contact MP**1** is limited, so the first conductive layer M**1** cannot be directly formed on the gate contact MP**1**, otherwise the first conductive layer M**1** will directly contact other adjacent contact structures such as the source.
[0029] In order to solve the above problems, in this embodiment, in addition to forming the gate contact MP**1**, a via structure V**0** is additionally formed on the gate contact MP**1** and the source/drain contacts MD**1**, and then a first conductive layer M**1** is formed above the via structure V**0**. The materials of the via structure V**0** and the first conductive layer M**1** described here are, for example, metals such as tungsten, cobalt, copper, aluminum, gold, silver, etc., but are not limited thereto. The via structure V**0** and the first conductive layer M**1** are used to connect the lower high voltage transistor element to other elements formed above. That is, in order to overcome the problem of insufficient height of the gate contact MP**1**, the original single contact structure is

replaced by two overlapping contact structures (namely, the gate contact MP**1** and the via structure V**0**). In addition, the via structure V**0** and the first conductive layer M**1** are located in a dielectric layer **14**, for example, an ultra-low dielectric constant (ULK) layer, and its dielectric constant is preferably lower than 2.9, but not limited to this. Generally, the commonly used ULK materials may include Black Diamond (low dielectric constant material of carbon-doped silicon oxide introduced by Applied Materials Company), MSQ (methylsilsesquioxane), porous SiLK (a low dielectric constant material developed by Dow Chemical), etc., but are not limited to this.
[0030] However, the height of the via structure V**0** is also limited by the process. For example, in this embodiment, the height of the via structure V**0** is about 480 angstroms, and the first conductive layer M**1** is formed on the via structure V**0**, and a part of the first conductive layer M**1** extends laterally from the medium/high voltage device region **10** into the low voltage device region **20**, so the distance between the bottom surface BS**1** of the first conductive layer M**1** in the low voltage device region **20** and the top surface TS**1** of the source/drain contact MD**1** below is only about 480 angstroms (same as the height of via structure V**0**). In this case, the applicant found another problem, that is, when the medium/high voltage semiconductor device is operating, it may produce a time dependent dielectric breakdown (TDDB) effect, that is, the current may penetrate through the dielectric layer **14** and flow to the source/drain contacts MD below, resulting in that the circuit cannot be connected to the expected device. For example, in this embodiment, as shown in FIG. **2**, if the source/drain contacts MD**2** in the low voltage device region **20** is connected to the gate contact MP**2**, then the current may flow from the medium/high voltage semiconductor device in the medium/high voltage device region **10** to the device in the low voltage device region **20** and then to another medium/high voltage semiconductor device in the medium/high voltage device region **30** through the path P, so the medium/high voltage semiconductor devices in two different regions will be influenced by each other, causing electrical errors and device damage.
[0031] It should be noted that the device pattern located in the low voltage device region **20** in FIG. **1** of the present invention may be changed as required, and is not limited to the structure shown in FIG. **1**. Although the source/drain contacts MD**2** are designed to be connected to the gate contact MP**2** in FIG. **1**, this pattern structure is only one example of the present invention. In other embodiments of the present invention, other devices such as transistors may be formed in the low voltage device region **20**, and the source/drain contacts MD**2** and the gate contact MP**2** are not in contact with each other. However, even if the source/drain contacts MD**2** and the gate contact MP**2** are not in contact with each other, the current in the medium/high voltage device region **10** may still pass through the first conductive layer M**1** and penetrate the dielectric layer **14** and then be transmitted to the devices in the low voltage device region **20**, and the device quality in the low voltage device region will be affected.
[0032] The applicant found that the structure of FIG. **2** has the above-mentioned probability of possible defects, especially at the boundary of the medium/high voltage device region and the low voltage device region. Therefore, in order to improve the structure of FIG. **2**, the applicant provides another structure. Please refer to FIG. **3** and FIG. **4**. FIG. **3** shows a schematic top view near a boundary between a medium/high voltage device region and a low-voltage device region in a second embodiment of the present invention, and FIG. **4** is a schematic cross-sectional view of the medium/high voltage device region and the low voltage device region in the second embodiment of the present invention, especially the cross-sectional structure taken along section line B-B′in FIG. **3**. In this embodiment, since it is known that the bottom surface BS**1** of the first conductive layer M**1** is too close to the top surface TS**1** of the source/drain contact MD below, the problem of time dependent dielectric breakdown (TDDB) may occur, in this embodiment, the first conductive layer M**1** is not extended to the low voltage device region **20**. The via structure V**1** and the second conductive layer M**2** are formed on the first conductive layer M**1**. If the device needs to extend into the low voltage device region **20** in configuration, the second conductive layer M**2** extends into the range of the low voltage device region **20**. In other words, the distance between the bottom surface

BS**2** of the second conductive layer M**2** and the top surface TS**1** of the second source/drain contacts MD**2** below is far enough, at least more than 700 angstroms, so it is not easy to cause the problem that the above-mentioned time dependent dielectric breakdown causes the current to flow to the devices in the low voltage device region **20** unexpectedly. It can effectively solve the problem of time dependent dielectric breakdown at the boundary of medium/high voltage device region and low voltage device region under the process of 17 nm, and improve the quality of the device.

[0033] Based on the above description and drawings, the present invention provides a semiconductor structure, which comprises a substrate **1**, on which a first medium/high voltage device region **10** and a low voltage device region **20** are defined adjacent to each other, the first medium/high voltage device region **10** comprises a first gate contact MP**1** and first source/drain contacts MD**1**, and the low voltage device region **20** comprises two second source/drain contacts MD**2** and a second gate contact MP**2**. The second gate contact MP**2** is located between and directly contacts two second source/drain contacts MD**2**, a first conductive layer M**1** is located in the first medium/high voltage device region **10** and electrically connected to the first gate contact MP**1** or the first source/drain contact MD**1**, wherein the first conductive layer M**1** does not extend into the low voltage device region **20**, a second conductive layer M**2** located above the first conductive layer M**1** and spanning the first medium/high voltage device region **10** and the low voltage device region **20**, a the second conductive layer M**2** is electrically connected with the first conductive layer M**1**.

[0034] In some embodiments of the present invention, the first medium/high voltage device region **10** further includes a first gate structure G**1** spanning a diffusion region D, wherein the first gate contact MP**1** is electrically connected to the first gate structure G**1**, and the first source/drain contact MD**1** is electrically connected to the diffusion region D.

[0035] In some embodiments of the present invention, the low voltage device region **20** further includes a second gate structure G**2** spanning a plurality of fin structures F, and the second source/drain contacts MD**2** are electrically connected to the plurality of fin structures F.

[0036] In some embodiments of the present invention, a top surface of the second gate contact MP**2** is aligned with the top surfaces of the two second source/drain contacts MD**2**.

[0037] In some embodiments of the present invention, in the low voltage device region **20**, the conductive material layer is not included at the same level as the first conductive layer M**1** (which means that the first conductive layer M**1** will not extend to the low voltage device region **20**, and other wire/conductive materials will not be included in the low voltage device region **20** at the same level as the first conductive layer M**1**).

[0038] In some embodiments of the present invention, the second conductive layer M**2** extends into the low voltage device region **20** and is located directly above at least one second source/drain contact MD**2**.

[0039] In some embodiments of the present invention, in the low voltage device region **20**, the distance between a bottom surface BS**2** of the second conductive line layer M**2** and a top surface TS**1** of the second source/drain contact MD**2** in a vertical direction is greater than **700** angstroms (according to the applicant's experiment, time dependent dielectric breakdown can be avoided when the distance is greater than **700** angstroms).

[0040] In some embodiments of the present invention, a second medium/high voltage device region **30** is further included, wherein the low voltage device region **20** is located between the first medium/high voltage device region **10** and the second medium/high voltage device region **30**, and the low voltage device region **20** is directly adjacent to the first medium/high voltage device region **10** and the second medium/high voltage device region **30**.

[0041] In some embodiments of the present invention, the elements comprised in the second medium/high voltage device region **30** and the elements comprised in the first medium/high voltage device region **10** are arranged in mirror images (which mean the symmetrical arrangement of these two patterns).

[0042] In some embodiments of the present invention, the first source/drain contact MD**1** in the first medium/high voltage device region **10** is directly adjacent to one of the second source/drain contacts MD**2** in the low voltage device region **20** (that is, no other elements are included between the first source/drain contact MD**1** and the second source/drain contact MD**2** on both sides of the boundary line B**1** in cross section).
[0043] The invention also provides a method for manufacturing a semiconductor structure, which comprises providing a substrate **1**, wherein a first medium/high voltage device region **10** and a low voltage device region **20** are defined on the substrate **1**, the first medium/high voltage device region **10** comprises a first gate contact MP**1** and a first source/drain contact MD**1**, and the low voltage device region **20** comprises two second source/drain contacts MD**2** and a second gate contact MP**2**. The second gate contact MP**2** is located between two second source/drain contacts MD**2**, and directly contacts the two second source/drain contacts MD**2** to form a first conductive layer M**1** located in the first medium/high voltage device region **10** and electrically connected with the first gate G**1** contact or the first source/drain contact MD**1**, wherein the first conductive layer M**1** does not extend into the low voltage device region **20**, and a second conductive layer M**2** is formed on the first conductive layer M**1**, and the second conductive layer M**2** spans the first medium/high voltage device region **10** and the low voltage device region **20**, wherein the second conductive layer M**2** is electrically connected with the first conductive layer M**1**.
[0044] To sum up, the applicant found that with the development of technology, the size of semiconductor devices is getting smaller and smaller, especially at the boundary of medium/high voltage device regions and low voltage device regions, various structural problems are easy to occur. For example, in the conventional 22 nm or 28 nm process, time dependent dielectric breakdown will not occur at the boundary of the medium/high voltage device region and the low voltage device region. However, as the process progresses below 17 nm, the conductive layer in the medium/high voltage device region is too close to the device in the low voltage device region, which may affect the quality of other devices due to time dependent dielectric breakdown. Therefore, the present invention aims at solving this problem, and specifically designs a semiconductor structure so that the first conductive layer in the medium/high voltage device region does not extend to the low voltage device region, and the second conductive layer in the medium/high voltage device region extends to the low voltage device region. Therefore, without changing the process conditions and being limited by the process, the problem of time dependent dielectric breakdown caused by the short distance between the conductive layer in the medium/high voltage device region and the device in the low voltage device region can be avoided, and the quality of the device can be improved.
[0045] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the invention. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

Claims

1. A semiconductor structure comprising: a substrate, on which a first medium/high voltage device region and a low voltage device region are defined adjacent to each other, wherein the first medium/high voltage device region comprises a first gate contact and a first source/drain contact, and the low voltage device region comprises two second source/drain contacts and a second gate contact, wherein the second gate contact is located between the two second source/drain contacts and directly contacts the two second source/drain contacts; a first conductive layer located in the first medium/high voltage device region and electrically connected to the first gate contact or the first source/drain contact, wherein the first conductive layer does not extend into the low voltage device region; and a second conductive layer located above the first conductive layer and spanning

the first medium/high voltage device region and the low voltage device region, wherein the second conductive layer is electrically connected with the first conductive layer.
2. The semiconductor structure according to claim 1, wherein the first medium/high voltage device region further comprises a first gate structure spanning a diffusion region, wherein the first gate contact is electrically connected to the first gate structure, and the first source/drain contact is electrically connected to the diffusion region.
3. The semiconductor structure according to claim 1, wherein the low voltage device region further comprises a second gate structure spanning a plurality of fin structures, and the second source/drain contact is electrically connected with the plurality of fin structures.
4. The semiconductor structure according to claim 1, wherein a top surface of the second gate contact is aligned with the top surfaces of the two second source/drain contacts.
5. The semiconductor structure according to claim 1, wherein a conductive material layer is not included in the low voltage device region at the same level as the first conductive layer.
6. The semiconductor structure according to claim 1, wherein the second conductive layer extends into the low voltage device region and is located directly above at least one second source/drain contact.
7. The semiconductor structure according to claim 6, wherein the distance between a bottom surface of the second conductive layer and a top surface of the second source/drain in a vertical direction is greater than 700 angstroms in the low voltage device region.
8. The semiconductor structure according to claim 1, further comprising a second medium/high voltage device region, wherein the low voltage device region is located between the first medium/high voltage device region and the second medium/high voltage device region, and the low voltage device region is directly adjacent to the first medium/high voltage device region and the second medium/high voltage device region.
9. The semiconductor structure according to claim 8, wherein the devices comprised in the second medium/high voltage device region and the devices comprised in the first medium/high voltage device region are arranged in mirror images with each other.
10. The semiconductor structure according to claim 1, wherein the first source/drain contact in the first medium/high voltage device region is directly adjacent to one of the second source/drain contacts in the low voltage device region when viewed from a cross section.
11. A manufacturing method of a semiconductor structure, comprising: providing a substrate on which a first medium/high voltage device region and a low voltage device region are defined adjacent to each other, wherein the first medium/high voltage device region comprises a first gate contact and a first source/drain contact, and the low voltage device region comprises two second source/drain contacts and a second gate contact, wherein the second gate contact is located between the two second source/drain contacts and directly contacts the two second source/drain contacts; forming a first conductive layer located in the first medium/high voltage device region and electrically connected to the first gate contact or the first source/drain contact, wherein the first conductive layer does not extend into the low voltage device region; and forming a second conductive layer above the first conductive layer and spanning the first medium/high voltage device region and the low voltage device region, wherein the second conductive layer is electrically connected with the first conductive layer.
12. The method for manufacturing a semiconductor structure according to claim 11, wherein the first medium/high voltage device region further comprises a first gate structure spanning a diffusion region, wherein the first gate contact is electrically connected to the first gate structure, and the first source/drain contact is electrically connected to the diffusion region.
13. The method for manufacturing a semiconductor structure according to claim 11, wherein the low voltage device region further comprises a second gate structure spanning a plurality of fin structures, and the second source/drain contact is electrically connected to the plurality of fin structures.

14. The method for manufacturing a semiconductor structure according to claim 11, wherein a top surface of the second gate contact is aligned with the top surfaces of the two second source/drain contacts.

15. The method for manufacturing a semiconductor structure according to claim 11, wherein a conductive material layer is not included at the same level as the first conductive layer in the low voltage device region.

16. The manufacturing method of a semiconductor structure according to claim 11, wherein the second conductive layer extends into the low voltage device region and is located directly above at least one second source/drain contact.

17. The method for manufacturing a semiconductor structure according to claim 16, wherein the distance between a bottom surface of the second conductive layer and a top surface of the second source/drain in a vertical direction is greater than **700** angstroms in the low voltage device region.

18. The method for manufacturing a semiconductor structure according to claim 11, further comprising defining a second medium/high voltage device region, wherein the low voltage device region is located between the first medium/high voltage device region and the second medium/high voltage device region, and the low voltage device region is directly adjacent to the first medium/high voltage device region and the second medium/high voltage device region.

19. The method for manufacturing a semiconductor structure according to claim 18, wherein the devices comprised in the second medium/high voltage device region and the devices comprised in the first medium/high voltage device region are arranged in mirror images.

20. The method for manufacturing a semiconductor structure according to claim 11, wherein the first source/drain contact in the first medium/high voltage device region is directly adjacent to one of the second source/drain contacts in the low voltage device region when viewed from a cross section.